Macroeconomic Forecasting with Time-Series Foundation Models

A Reproduction and Extension of Carriero, Pettenuzzo & Shekhar (arXiv:2407.00890)

Project Report | October 2026

Abstract

Time-series foundation models (TSFMs) promise accurate zero-shot forecasts without domain-specific training. Carriero, Pettenuzzo and Shekhar test this promise on the FRED-MD macroeconomic database and conclude that only a few TSFMs beat a simple autoregression, that these are broadly comparable to but less stable than Bayesian VARs (BVARs) and factor models, and that a hybrid in which a TSFM corrects BVAR residuals performs best. This project reproduces and extends that study on 60 FRED-MD series (1959–2022) with an expanding-window design of 48 quarterly origins (2010 Q1–2022 Q4) and horizons of 1, 3, 6 and 12 months. We compare AR(1), a Minnesota BVAR and a Factor-AR with six zero-shot TSFMs (TimesFM, TimeGPT, Chronos-Bolt, MOIRAI, TTM, Lag-Llama) and a BVAR+TimesFM hybrid, using median RMSFE relative to AR(1). TimesFM, TimeGPT and Chronos-Bolt edge below the benchmark at short horizons (0.983–0.996 at h = 1), consistent with the paper's finding that zero-shot TSFMs are competitive but not dominant. Our BVAR and Factor-AR, however, underperform AR(1), contradicting the paper, and the hybrid gain is marginal. We discuss probable causes and the steps needed to close the gap.

Keywords: macroeconomic forecasting; foundation models; FRED-MD; Bayesian VAR; zero-shot forecasting; relative RMSFE

1. Workflow

Figure 1 shows the end-to-end pipeline in five stages: (i) data preparation, (ii) an expanding-window forecasting engine, (iii) two parallel model families, (iv) a hybrid residual-correction model, and (v) evaluation against the AR(1) benchmark.

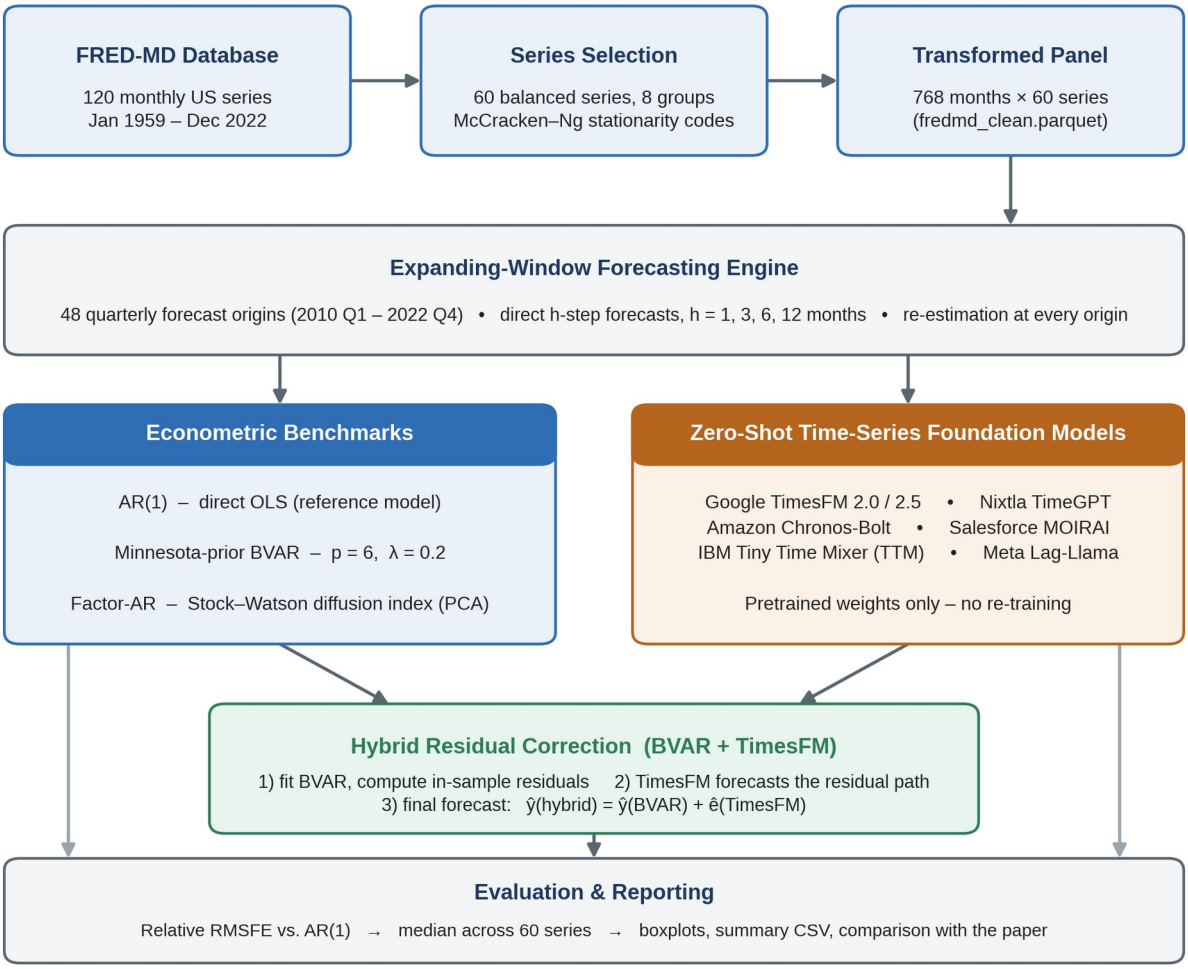


Figure 1. End-to-end workflow of the forecasting and evaluation pipeline.

2. Methodology

2.1 Data and preprocessing

We use 60 balanced monthly series from FRED-MD (McCracken and Ng, 2016), January 1959 to December 2022, spread across all eight standard groups: output and income (8), labour market (16), housing (5), consumption and orders (5), money and credit (6), interest and exchange rates (11), prices (7) and stock market (2). Each series is made stationary with its McCracken–Ng transformation code (levels, first or second differences, log or log-differences), giving a 768 × 60 panel.

2.2 Forecasting design

Forecasts are produced with an expanding window. At each of 48 quarterly origins from 2010 Q1 to 2022 Q4, every model uses all data available up to the origin and forecasts h = 1, 3, 6 and 12 months ahead. The econometric models are re-estimated at every origin; the TSFMs receive the history up to the origin as context and are never re-trained.

2.3 Econometric benchmarks

AR(1): direct OLS forecasts for each series and horizon; it is the reference for all ratios. **BVAR:** a Minnesota-prior VAR with p = 6 lags and overall tightness λ = 0.2; with 60 variables each equation has 361 coefficients, so shrinkage is essential. **Factor-AR:** a Stock–Watson diffusion-index model in which principal-component factors augment an autoregression of the target series.

2.4 Zero-shot foundation models

Six pretrained TSFMs are applied without any macroeconomic fine-tuning: Google TimesFM (2.0/2.5), Nixtla TimeGPT (API), Amazon Chronos-Bolt, Salesforce MOIRAI, IBM Tiny Time Mixer (TTM) and Meta Lag-Llama. They convert a series into patches or quantised tokens, process them with a transformer (TTM uses an MLP-mixer) and return a multi-step forecast. As the paper stresses, pretraining corpora are not time-stamped and some include FRED-MD, so a strictly out-of-sample exercise cannot be guaranteed.

2.5 Hybrid residual-correction model

Following the paper, a BVAR is fitted first and its in-sample residuals are computed. TimesFM then forecasts the residual path and the correction is added to the BVAR forecast:

$$\hat{y}_{t+h}^{\text{hybrid}} = \hat{y}_{t+h}^{\text{BVAR}} + \hat{e}_{t+h}^{\text{TSFM}}$$

2.6 Evaluation metric

Accuracy is measured by the root mean squared forecast error relative to AR(1), computed for each series j and horizon h; values below one mean the model beats AR(1). We report the median across series.

$$\text{RMSFE}_{j,h}^{\text{rel}} = \sqrt{\frac{\sum_{\tau}\left(y_{j,\tau+h} - \hat{y}_{j,\tau+h}^{\text{model}}\right)^2}{\sum_{\tau}\left(y_{j,\tau+h} - \hat{y}_{j,\tau+h}^{\text{AR(1)}}\right)^2}}$$

3. Our Approach versus the Original Paper

The project follows the paper's logic: same dataset, same relative-RMSFE metric, same family of benchmarks and the same BVAR-plus-TSFM hybrid. It differs in scale and in several design choices, summarised in Table 1. Notably, we add Chronos-Bolt and newer TimesFM versions, while the paper's additional nonlinear benchmarks (NNAR, BART), significance tests and fine-tuning experiments are not yet replicated.

Table 1. Comparison of the original study and this work.

Aspect	Original paper (Carriero et al.)	This work
Data	FRED-MD, 120 monthly series, 1959–2022	FRED-MD, 60 balanced series, 1959–2022
Evaluation	Jan 1985 – Dec 2019 (main), 2020–2022 separately; monthly origins; h = 1, …, 12	2010 Q1 – 2022 Q4; 48 quarterly origins; h = 1, 3, 6, 12
Benchmarks	AR(1), BVAR, Factor model, NNAR, BART	AR(1), BVAR, Factor-AR
BVAR prior	Minnesota + sum-of-coefficients + single-unit-root; p = 6; hyper-parameters by marginal likelihood	Minnesota prior; p = 6; λ = 0.2 fixed
TSFMs	MOIRAI (base, large), TimesFM, TTM, TimeGPT, Lag-Llama	TimesFM 2.0/2.5, TimeGPT, Chronos-Bolt, MOIRAI, TTM, Lag-Llama
Hybrid	BVAR + MOIRAI-large and BVAR + TimesFM	BVAR + TimesFM
Extra analyses	Diebold–Mariano tests, COVID split, data-leakage test, persistence split, fine-tuning	Not yet implemented

4. Results and Comparison with the Paper

Table 2 reports the results of this work and Table 3 those of the original paper, each covering the single models and the hybrid models. All entries are the median RMSFE relative to AR(1) across series (values below 1, in bold, beat AR(1)). Because the two studies use different series sets, evaluation windows and model versions, the comparison is about patterns rather than exact decimals. Figure 2 contrasts the shortest and longest horizons.

Table 2. Results of this work: median relative RMSFE (60 FRED-MD series, 2010 Q1 – 2022 Q4, 48 quarterly origins).

Model	h = 1	h = 3	h = 6	h = 12
Single models				
AR(1) (benchmark)	1.000	1.000	1.000	1.000
BVAR (Minnesota)	1.332	1.262	1.167	1.108
Factor-AR	1.507	1.192	1.089	1.029
TimesFM 2.0/2.5	**0.983**	**0.995**	**0.997**	1.000
TimeGPT	**0.987**	**0.997**	**0.994**	**0.999**
Chronos-Bolt	**0.996**	**0.997**	**0.998**	1.002
MOIRAI	1.031	1.004	1.003	1.013
TTM	1.117	1.040	1.019	1.007
Lag-Llama	1.043	1.022	1.020	1.014
Hybrid model				
BVAR + TimesFM	1.331	1.250	1.168	1.111

Chronos-Bolt was evaluated only in this work. The hybrid combines the BVAR with TimesFM residual forecasts.

Table 3. Results of the original paper: median relative RMSFE (120 FRED-MD series, Jan 1985 – Dec 2019, monthly origins).

Model	h = 1	h = 3	h = 6	h = 12
Single models – econometric				
AR(1) (benchmark)	1.000	1.000	1.000	1.000
BVAR	**0.983**	**0.970**	**0.991**	**0.999**
Factor model	**0.965**	**0.980**	**0.992**	**0.999**
NNAR	**0.988**	**0.992**	**0.999**	1.000
BART	1.017	1.024	1.011	1.023
Single models – zero-shot TSLMs				
MOIRAI-base	1.008	**0.973**	**0.990**	**0.999**
MOIRAI-large	**0.999**	**0.978**	**0.991**	1.001
TimesFM	1.014	**0.980**	**0.990**	1.001
TimeGPT	1.077	1.044	1.025	1.063
TTM-Enhanced	1.044	**0.993**	**0.995**	1.002
Lag-Llama	1.015	**0.997**	1.002	1.009
Hybrid models				
BVAR + MOIRAI-large	**0.973**	**0.960**	**0.983**	**0.998**
BVAR + TimesFM	**0.980**	**0.956**	**0.984**	**0.999**

Source: arXiv:2407.00890v4, Tables 2–4. The repository cites a later revision, so figures may differ slightly. In Figure 2 and in the text, MOIRAI refers to MOIRAI-large, TTM to TTM-Enhanced and the hybrid to the BVAR + TimesFM version.

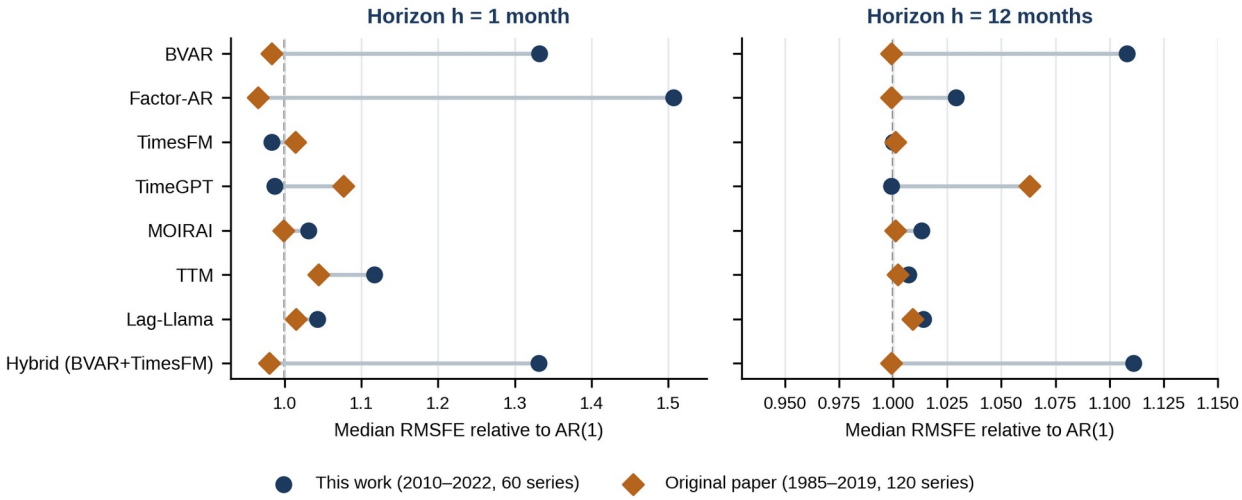


Figure 2. Median relative RMSFE, this work versus the original paper (dashed line = AR(1) benchmark).

Where the two studies agree

Zero-shot TSFMs are competitive but not dominant. TimesFM stays within about 2% of AR(1) in both studies (0.983–1.000 here, 0.980–1.014 in the paper), and nearly all TSFMs drift to roughly 1.00 at h = 12 (TimeGPT is the exception in the paper), i.e. long-horizon forecasts are no better than the benchmark. **The hybrid helps at short horizons.** Adding TimesFM residual forecasts lowers the BVAR ratio at h = 1 and 3 (by 0.001 and 0.012 here; 0.003 and 0.014 in the paper), but the gains are small in both studies.

Where they disagree

Econometric models (main discrepancy). The paper finds BVAR and Factor models beat AR(1) (0.965–0.999). Our BVAR and Factor-AR are worse than AR(1) at every horizon (1.332 and 1.507 at h = 1), approaching parity only at h = 12. Probable contributors, not yet isolated, are the fixed prior tightness ($\lambda = 0.2$) in a 60-variable system instead of marginal-likelihood tuning with sum-of-coefficients and unit-root priors, the Factor-AR lag choice, and an evaluation window that includes the pandemic years, which the paper also reports as difficult for the factor model. This should be resolved before drawing conclusions about the econometric side. **Ranking of TSFMs.** TimeGPT is the weakest TSFM in the paper (1.077 at h = 1) but one of the best here (0.987), whereas MOIRAI is the paper's best model but is mediocre here (1.031). Different model versions, series sets and sample periods are plausible reasons. **Hybrid at long horizons.** The paper's hybrid matches or improves on the BVAR at all horizons; ours is flat or marginally worse at h = 6 and 12 (1.168 and 1.111 versus the BVAR's 1.167 and 1.108).

5. Conclusion

We implemented an end-to-end replication of a recent study on foundation models for macroeconomic forecasting and extended it with Chronos-Bolt and newer TimesFM versions. The qualitative conclusions on TSFMs carry over: zero-shot models are roughly on par with a simple AR(1), gain little at short horizons and none at h = 12, and a BVAR-plus-TSFM hybrid yields at best small improvements. The key discrepancy is that our BVAR and Factor-AR do not beat AR(1), which suggests that the benchmark implementation, rather than the foundation models, is the main item to verify. Differences in sample, series coverage and model versions also limit direct numerical comparison. Until the benchmarks are aligned and significance tests are added, the results should be read as a faithful partial replication, not as evidence against the paper.

6. Future Work

- **Fix and validate the benchmarks:** tune BVAR hyper-parameters by marginal likelihood, add sum-of-coefficients and single-unit-root priors, and select Factor-AR lags by BIC.
- **Align the design with the paper:** 120 series, monthly origins, horizons 1–12, 1985–2019 main window, with 2020–2022 reported separately.
- **Add statistical testing and tail analysis:** Diebold–Mariano tests and per-series dispersion (min, max, standard deviation) to check whether the hybrid reduces extreme errors.
- **Study data leakage:** re-estimate the econometric models on the full sample and check pretraining overlap with FRED-MD for the newer TSFM versions.
- **Extend the models:** parameter-efficient fine-tuning, hybrids with MOIRAI and Chronos-Bolt, and NNAR and BART baselines.
- **Move toward real-time use:** FRED-MD vintages, density forecasts, and additional (e.g. text-based) signals.

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